

L5a: Network-Flow Models

Capacity, conservation, and maximum flow
CHEME 5800 Cornell University

Objectives for Today

How much flow can a capacity-limited network deliver, and how can we show that this amount is the maximum?

Today's concepts

- **Formulate the problem:** define capacities, conservation, and net source outflow.
- **Apply Ford–Fulkerson:** construct residual graphs, revise earlier routes, and choose augmenting paths.
- **Verify maximum flow:** check feasibility and use a cut capacity to prove that more flow is impossible.

Lecture: Network-flow models: capacity, conservation, and maximum flow

Lecture and Example

Lecture: Network-flow models

Formulate a flow problem, increase flow through residual paths, and establish a bound on what the network can deliver.

Example: Maximum flow in a worker–task network

How many assignments can the workers complete? Build the network, compute flows using two path-search rules, and check the results.

One unit of flow represents **one completed assignment**.

Formulating a Maximum-Flow Problem

Represent the application as a finite directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with vertices $\mathcal{V}$ and edges $\mathcal{E}$.

- The **source** s supplies flow; the **sink** t receives it, with $s \neq t$.
- Each original edge (u, v) has a finite **capacity** $c(u, v) \geq 0$.
- The **flow** $f(u, v)$ is the amount assigned to that edge, in the same units as capacity.

Worker availability and task limits become capacities. Data transmission and material transport use the same formulation.

Capacity and Conservation Define Feasibility

Each original edge's flow $f(u, v)$ must stay within its capacity $c(u, v)$:

$$\underbrace{0 \leq f(u, v) \leq c(u, v)}_{\text{capacity constraint}}, \quad (u, v) \in \mathcal{E}.$$

At an intermediate vertex $v \in \mathcal{V} \setminus \{s, t\}$, incoming and outgoing flow must balance:

$$\underbrace{\sum_{u:(u,v) \in \mathcal{E}} f(u, v)}_{\text{inflow}} = \underbrace{\sum_{w:(v,w) \in \mathcal{E}} f(v, w)}_{\text{outflow}}.$$

Intermediate vertices neither create nor consume flow. A **feasible flow** satisfies both constraints.

Maximize Net Flow Leaving the Source

The flow value $|f|$ measures net outflow from the source s :

$$|f| = \underbrace{\sum_{v:(s,v) \in \mathcal{E}} f(s,v)}_{\text{leaving } s} - \underbrace{\sum_{u:(u,s) \in \mathcal{E}} f(u,s)}_{\text{entering } s}.$$

Subtract returning flow. Conservation makes $|f|$ equal to the net inflow at the sink t .

The maximum-flow problem seeks a feasible flow f^* with largest value:

$$|f^*| = \max_{f \text{ feasible}} |f|.$$

Worker-task model: no edges enter the source, so its total outflow counts completed assignments.

Ford–Fulkerson Method

Start with zero flow. Repeatedly find a source-to-sink route that allows the total flow to increase.

A **residual graph** records changes we can make to the current flow:

- **Forward edges** record unused capacity.
- **Backward edges** record flow we can cancel to revise an earlier routing choice.

Ford–Fulkerson is a **method**: it leaves the path-selection rule open.

What Is a Residual Graph?

Suppose the original edge (u, v) has capacity $c(u, v) = 5$ and currently carries flow $f(u, v) = 3$.

Forward: add flow

Unused capacity from u to v :

$$c(u, v) - f(u, v) = 2.$$

Backward: cancel flow

Flow we can cancel using $v \rightarrow u$:

$$f(u, v) = 3.$$

A backward residual edge is bookkeeping. It reduces an existing flow; it does not require a physical reverse connection.

Defining Residual Capacities

Assume no self-loops or oppositely directed pairs of original edges. For an ordered pair (u, v) , the residual capacity is:

$$\text{residual}(u, v) = \begin{cases} c(u, v) - f(u, v), & (u, v) \in \mathcal{E}, \\ f(v, u), & (v, u) \in \mathcal{E}, \\ 0, & \text{otherwise.} \end{cases}$$

The residual graph $\mathcal{G}_f = (\mathcal{V}, \mathcal{E}_f)$ retains all vertices and includes only edges with positive residual capacity:

$$\mathcal{E}_f = \{(u, v) \in \mathcal{V} \times \mathcal{V} : \text{residual}(u, v) > 0\}.$$

Scope: the worker–task network satisfies these assumptions; f remains nonnegative on original edges.

How Do We Augment the Flow?

An **augmenting path** P is a directed path from s to t in the residual graph. Every edge on it has positive residual capacity.

The largest increase Δ along the chosen path is its **bottleneck capacity**:

$$\Delta = \min_{(u,v) \in P} \text{residual}(u,v).$$

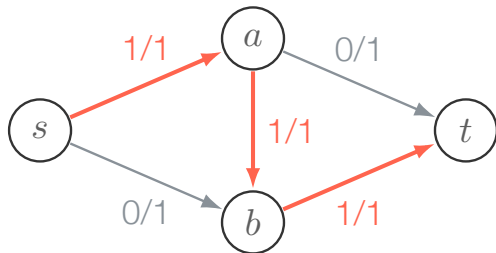
- On a forward edge, Δ limits how much flow we can add.
- On a backward edge, Δ limits how much flow we can cancel.

Use the same Δ on every path edge to preserve conservation.

A Reverse Edge Revises an Earlier Route

All five original edges have capacity one.

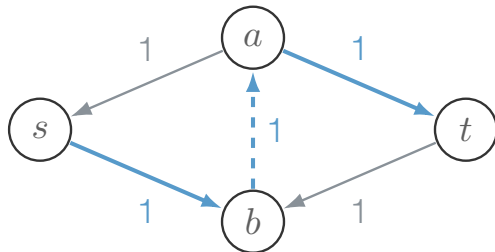
First flow: $|f| = 1$



Labels: flow / capacity

$s \rightarrow a \rightarrow b \rightarrow t$

Residual path: $\Delta = 1$



Labels: residual capacity

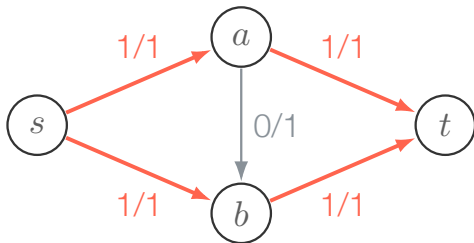
$s \rightarrow b \rightarrow a \rightarrow t$

The dashed step $b \rightarrow a$ cancels the earlier flow on (a, b) .

Rerouting Raises the Flow Value to Two

Augment along $s \rightarrow b \rightarrow a \rightarrow t$ with $\Delta = 1$.

Revised flow: $|f| = 2$



Labels: flow / capacity

$s \rightarrow a \rightarrow t$ and $s \rightarrow b \rightarrow t$

Original edge updates

Edge	Before	After
(s, b)	0	1
(a, b)	1	0
(a, t)	0	1

Other edge flows stay at one; conservation is preserved.

Ford–Fulkerson Procedure

Initialize $f(u, v) = 0$ for every original edge. Then repeat:

1. Construct the residual graph and find a path P from s to t . If no path exists, stop.
2. Find the bottleneck Δ along P .
3. Add Δ to each forward edge's flow; subtract Δ from the original flow corresponding to each backward edge.
4. Recompute residual capacities before searching again.

Each augmentation preserves feasibility and increases $|f|$ by Δ .

Edmonds–Karp Uses Breadth-First Search

Search the residual graph using **breadth-first search**, following only edges with positive residual capacity.

- Choose an augmenting path with the fewest edges.
- Neighbor order breaks ties between equal-length paths.
- Update residual capacities and search again after each augmentation.

With adjacency lists and exact arithmetic, the worst-case running time is:

$$\mathcal{O}(|\mathcal{V}| |\mathcal{E}|^2).$$

The bound depends on graph size, not capacity values.

Reference: [Sedgewick and Wayne, Algorithms: Maximum Flow](#)

Depth-First Search and Termination

Depth-first search finds a residual path; it need not have the fewest edges.

For nonnegative integer capacities and zero initial flow:

- Each augmentation adds at least one unit of flow.
- Finite source capacity bounds the total flow.
- The method terminates; the iteration count can depend on capacities.

Irrational capacities can allow a sequence of path choices that never terminates.

Compare Both Searches on the Same Network

The [worker-task example](#) applies two path-selection rules to the same capacities.

Course solver	Residual-path selection
Ford–Fulkerson implementation	Depth-first search
Edmonds–Karp	Breadth-first search; fewest edges

Both methods should give the same maximum-flow value. Their individual worker-task assignments may differ.

We next check feasibility and establish that the value is maximum.

Verifying Feasibility and Maximum Flow

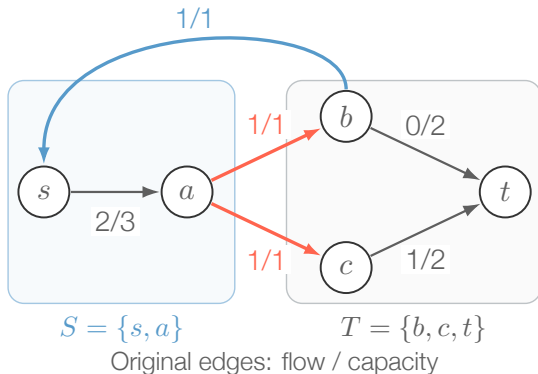
Independently check the reported edge flows:

1. **Capacities**: every original edge satisfies $0 \leq f(u, v) \leq c(u, v)$.
2. **Conservation**: each intermediate vertex has equal total inflow and outflow.
3. **Source–sink balance**: net source outflow equals net sink inflow.

These checks establish feasibility. To prove the flow is maximum, find an upper bound that the feasible flow reaches.

A Source–Sink Cut Gives an Upper Bound

A **source–sink cut** partitions $\mathcal{V}$ into disjoint sets S and T , with $s \in S$ and $t \in T$.



Cut capacity sums original capacities from S to T :

$$c(S, T) = \sum_{\substack{(u,v) \in \mathcal{E} \\ u \in S, v \in T}} c(u, v).$$

Here the two **outgoing cut edges** give $c(S, T) = 1 + 1 = 2$.

The **return edge** adds no cut capacity.

Cut Balance Bounds Every Feasible Flow

Sum net outflows over vertices in S . Internal edge flows cancel, and conservation leaves only the source's net outflow:

$$|f| = \underbrace{\sum_{\substack{(u,v) \in \mathcal{E} \\ u \in S, v \in T}} f(u,v)}_{S \rightarrow T} - \underbrace{\sum_{\substack{(u,v) \in \mathcal{E} \\ u \in T, v \in S}} f(u,v)}_{T \rightarrow S}.$$

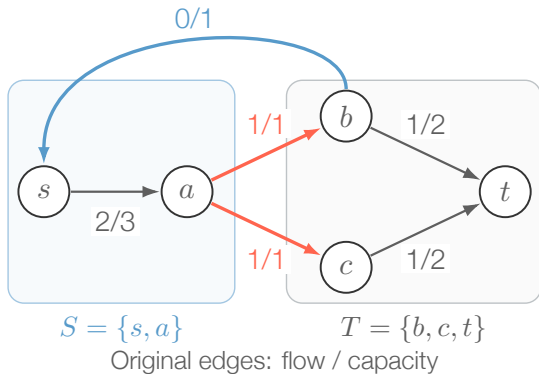
Forward flow is at most its capacity; return flow is nonnegative. Thus:

$$|f| \leq \sum_{\substack{(u,v) \in \mathcal{E} \\ u \in S, v \in T}} c(u,v) = c(S,T).$$

In the illustrated cut, $|f| = 2 - 1 = 1 \leq c(S,T) = 2$. The return flow reduces net delivery.

No Augmenting Path Identifies a Cut

When t is unreachable, let S contain the vertices reachable from s in the residual graph; $T = \mathcal{V} \setminus S$.



$S \rightarrow T$: **saturated**

Unused capacity would let the residual search enter T .

$T \rightarrow S$: **zero flow**

Positive flow would create a backward residual edge into T .

Residual augmentation $s \rightarrow b \rightarrow t$ cancels $b \rightarrow s$ and gives $|f| = 2$.

The Stopping Condition Proves Maximum Flow

Substitute saturated $S \rightarrow T$ edges and zero $T \rightarrow S$ flows into the cut balance:

$$|f| = c(S, T).$$

The feasible flow reaches a bound on every feasible flow. It is therefore maximum, and the cut has minimum capacity.

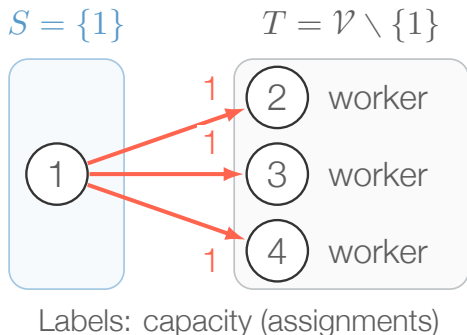
The **maximum-flow/minimum-cut theorem** states:

$$\max_{f \text{ feasible}} |f| = \min_{(S, T) \text{ source-sink cut}} c(S, T).$$

Reference: [David Karger, MIT 6.854: Flow](#)

Three Assignments Reach the Source-Cut Bound

The default worker–task network has three source-to-worker edges, each with capacity one assignment.



Put source 1 in S and every other vertex in T . The cut capacity is:

$$c(S, T) = 1 + 1 + 1 = 3.$$

The [worked example](#) attains this bound:

$$|f| = c(S, T) = 3.$$

Therefore, **three assignments** is the maximum.

Next: [L5b changes capacities](#) and examines how the maximum number of assignments changes.

Summary

We formulated, computed, and verified maximum flow.

Key takeaways

- **Capacity and conservation.** We maximized net source outflow subject to edge capacities and intermediate flow balances.
- **Residual augmentation.** We used bottlenecks to add or cancel flow while preserving feasibility. Edmonds–Karp chose paths with the fewest edges.
- **Proof of maximum flow.** We matched feasible flow to a cut bound, proving a maximum of three assignments in our example.

In [L5b](#), we examine how capacity changes affect this maximum.